NAME: _____

DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)

TASK: Perform Control Rod System Surveillance IAW S2.OP-ST.RCS-0001

TASK NUMBER: 50638

INITIAL CONDITIONS:

- Unit 2 is at 100% power BOL.
- No major equipment is out of service and no Tech Specs are active.
- The rod control system surveillance is in progress. All sections are complete, except for exercising Control Bank D.

INITIATING CUE:

- You are the Reactor Operator.
- The CRS directs you to complete the rod control system surveillance IAW S2.OP-ST.RCS-0001, Reactivity Control System Rod Control Assemblies.
- A Maintenance Technician is stationed at the Rod Control Power Cabinets (Relay Room)
- CRS directs that 15 steps of rod insertion will be performed to ensure each rod moves at least 10 steps.
- Notify the CRS when the testing is complete.
- Your evaluator will take care of all alarms not related to your task.

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
1. **Exercises Control Bank D at least 10 steps and upon completion of the test recognizes unexpected continuous rod movement requiring a manual reactor trip.**

NAME: _____

DATE: _____

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* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		♦ **ENSURE** marked up copy of S2.OP-ST.RCS-0001 is open and marked up on console. ♦ Provide copy of OP-AP-300-1001, PWR Control Rod Movement Requirements.			
		Operator reads and reviews procedures prior to start.	Operator reads P&Ls and reviews OP-AP-300-1001 prior to start		
	CUE:	Fill in the JPM Start Time when the student acknowledges the Initiating Cue. **START TIME**:_____			
	4.1.10.1	1. IF the reactor is critical, THEN **ENSURE** TAVG is within ±1°F of TREF.	Operator checks Tave/Tref recorder on 2RP4 and determines Tavg is within +/-1 F of Tref.		
*	4.1.10.2	2. **PLACE** Bank Selector Switch in the "CBD" position.	**Operator rotates selector switch clockwise to the CBD position**		

NAME: _____
DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)
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* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	4.1.10.3	3. **ENSURE** GRP. SELECT "B" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD.	Contacts Maint Technician at Power Cabinets**.** **CUE**: *Technician reports GRP SELECT 'B' lights are illuminated on Cabinets 21 BD and 22BD*.		
		Evaluators Note: Steps 4.1.10.4 and 4.1.10.5 will be performed three times consecutively prior to the 15 step insertion in step 6. **Evaluators Note:** IF the applicant moves the rods only one and a half a steps, THEN instruct the applicant to move the rods in the same direction then come back two full steps prior to reversing direction.			
	4.1.10.4	Insert Control Bank D 1 step.	**Operator inserts one step.**		
	4.1.10.5	Withdraw Control Bank D 1 step.	**Operator withdraws one step.**		
	4.1.10.4	Insert Control Bank D 1 step.	**Operator inserts one step.**		
	4.1.10.5	Withdraw Control Bank D 1 step.	**Operator withdraws one step.**		
	4.1.10.4	Insert Control Bank D 1 step.	**Operator inserts one step.**		
	4.1.10.5	Withdraw Control Bank D 1 step.	**Operator withdraws one step.**		

NAME: _____

DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)

TASK: Perform Control Rod System Surveillance IAW S2.OP-ST.RCS-0001

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	4.1.10.6	6. **MANEUVER** Control Bank **D** at least 10 steps in any one direction.	**Operator inserts Control Bank D 15 steps.**		
	4.1.10.7	7. **ENSURE** each rod in Control Bank **D** indicated rod movement of at least 10 steps.	**Evaluator's Note:** Operator may request to insert rods additional steps, IF so, just restate the request. Operator monitors rod position on P-250 and determines D bank rods all moved at least 10 steps. **Optional CUE**: *Reactor Engineer reports rods indicate 10 step movement.*		
*	4.1.10.8	8. **RECORD** CONTROL BANK **D** "Test Results" by initialing the SAT or UNSAT column using the Acceptance Criteria in Attachment 1, Rod Control Assembly Data.	**Operator records test results as SAT.**		
*	4.1.10.9	9. **RESTORE** Control Bank **D** to the pre-test position.	**Operator withdraws Bank D to previous position (ARO)**		
	4.1.11	**ALIGN** the Rod Control System as follows:			
		ALTERNATE PATH STARTS HERE:	**Continuous rod movement when rod bank is selected to Auto or Manual**		

NAME: _____

DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)

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* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	4.1.11.1	A. IF the reactor is critical, THEN **ENSURE** TAVG is within ±1°F of TREF.	Operator checks Tave/Tref recorder on 2RP4 and determines Tavg is within 1 F of Tref**.**		
	4.1.11.2	2. IF Turbine Power is ≤15%, THEN **PLACE** Bank Selector Switch in the "MAN" position. **Simulator Operator**: **ENSURE ET-1** is TRUE when rods selected to Auto. This will insert **MALF: RD0045, Uncontrolled Rod Insertion in AUTO AND MANUAL.**	N/A		

NAME: _____

DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)

TASK: Perform Control Rod System Surveillance IAW S2.OP-ST.RCS-0001

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	4.1.11.3	3. IF Turbine Power is >15%, THEN **PLACE** Bank Selector Switch in the "AUTO" OR "MAN" position as directed by the SM/CRS.	**CUE:** IF asked, *CRS directs rod bank selected to AUTO*. Operator determines that Rx power is > 15% and rotates selector switch counterclockwise to the AUTO position. Operator announces that rods are stepping in and no runback in progress. **CUE:** IF operator recommends to CRS to place rods in Manual, THEN state; ***understand placing rods to manual.*** Operator places rod bank switch to Manual and reports rod motion has NOT stopped. **Operator Manually trips the Reactor.** **JPM Complete once Reactor is Tripped.**		
		Examiner's Note: **The operator may refer to S2.OP-AB.ROD-0003, Continuous Rod Motion and take the actions in the AB to manually trip the reactor.**			

NAME: _____

DATE: _____

SYSTEM: Reactivity Control (SF-1) – Control Rod Drive System (CRDS)

TASK: Perform Control Rod System Surveillance IAW S2.OP-ST.RCS-0001

* #	STEP NO.	STEP (Shaded area denotes Critical Step) (* Critical Step) (# Sequential Critical Step)	STANDARD (Bolded area identifies Task Standard)	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	CUE:	**JPM is Complete** **RECORD** the STOP TIME. **STOP TIME**:_____	**Terminate the JPM when Reactor is tripped.**		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

JPM#: 20-01 NRC Sim-a

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

KH 1. Task description and number, JPM description and number are identified.

KH 2. Knowledge and Abilities (K/A) references are included.

KH 3. Performance location specified. (in-plant, control room, or simulator)

KH 4. Initial setup conditions are identified.

KH 5. Initiating and terminating cues are properly identified.

KH 6. Task standards identified and verified by SME review.

KH 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

KH 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. __25___ Date ____7-22-21_____

KH 9. Pilot test the JPM:
a. verify cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.

N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

KH 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: Date:

SME/Instructor: Date:

SME/Instructor: Date:

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INITIATING CUE:

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- Notify the CRS when the testing is complete.
- Your evaluator will take care of all alarms not related to your task.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION: SALEM

SYSTEM: Chemical and Volume Control System (CVCS)

TASK: Perform Manual Makeup to VCT IAW S2.OP-SO.CVC-0006

TASK NUMBER: 0040130101

JPM NUMBER: 20-01 NRC Sim-b

ALTERNATE PATH: [ ]

K/A NUMBER: 004 A4.04

IMPORTANCE FACTOR: 3.2 (**RO**) 3.6 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator / Perform

REFERENCES: S2.OP-SO.CVC-0006, Rev. 25 (checked 8-3-21)
S2.OP-AB.CVC-0001, Rev. 14
S2.RE-RA.ZZ-0012 Rev. 225

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 15 min

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By: **K.Hantho** **Instructor** **Date: 8-3-21**

Validated By: **Rydell/Zirkle** **SME or Instructor** **Date: 8-12-21**

Approved By: **M. Wadusky (signature on file)** **Training Department** **Date: 2-10-22**

Approved By: **W. Hargrave** **Operations Department** **Date: 1-11-22**

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:
GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

REVISION HISTORY

JPM NUMBER: 20-01 NRC Sim-b

Rev #	Date	Description	Validation Required
00	9-8-17	Added revision history and simulator setup pages. Editorial comments from IP 71111.11 FASA.	Yes
01	12-15-17	Added note for evaluator for step 5.2.4 that 2CV179 when placed in manual will initially go open from the closed position as per design.	No
02	8-3-21	Updated to current procedure revisions. JPM not affected by revisions.	Yes
03	1-3-22	Incorporated NRC comments from ES-301-7.	Yes

SIMULATOR SETUP INSTRUCTIONS

SYSTEM: Chemical and Volume Control System (CVCS)

TASK: Perform Manual Makeup to VCT IAW S2.OP-SO.CVC-0006

TASK NUMBER: 0040130101

SIMULATOR IC: IC-251

MALFUNCTIONS / REMOTES:

1. Reset the simulator to IC-251
2. The following malfunctions have been INSERTED:
   - **MALF CV0037 VCT LEVEL XMTR LT112 FAILS H/L, Final Value: 100.**
3. This completes the setup for this JPM.

OVERRIDES: None

SPECIAL INSTRUCTIONS:

- **Ensure the VCT level is about 30%.**
- **Ensure a marked up (partial procedure usage) of S2.OP-SO.CVC-0006 is available for candidate.**
- **Setup Plant computer on 2CC1 to CVCS system so operator can monitor VCT level using 2LT114.**